



Exercise 5: ASSOCIATION RULE MINING- APRIORI

Write a python program to find rules that describe associations by using Apriori algorithm

Steps in Apriori:

1. Set a minimum value for support and confidence. This means that we are only interested in finding rules for the items that have certain default existence (e.g. support) and have a minimum value for co-occurrence with other items (e.g. confidence).
2. Extract all the subsets having higher value of support than minimum threshold.
3. Select all the rules from the subsets with confidence value higher than minimum threshold.
4. Order the rules by descending order of Lift.

Example:

```
from apyori import apriori
```

```
transactions = [ ['beer', 'nuts'], ['beer', 'cheese'], ]
```

#CASE1:

```
results = list(apriori(transactions))
```

```
association_results = list(results)
```

```
print(results[0])
```

#CASE2:

```
min support=.5,minconfidence=.8
```

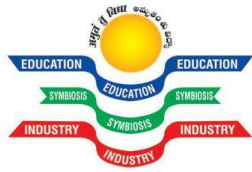
```
results = list(apriori(transactions,min_support=0.5, min_confidence=0.8))
```

```
association_results = list(results)
```

```
print(len(results))
```

```
print(association_results)
```

OUTPUT: 5



RelationRecord(items=frozenset({'beer'}), support=1.0,
ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=frozenset({'beer'}),
confidence=1.0, lift=1.0)])

Case 2:

3

[RelationRecord(items=frozenset({'beer'}), support=1.0,
ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=frozenset({'beer'}),
confidence=1.0, lift=1.0)]),

RelationRecord(items=frozenset({'cheese', 'beer'}), support=0.5,
ordered_statistics=[OrderedStatistic(items_base=frozenset({'cheese'}), items_add=frozenset({'beer'}),
confidence=1.0, lift=1.0)]),

RelationRecord(items=frozenset({'nuts', 'beer'}), support=0.5,
ordered_statistics=[OrderedStatistic(items_base=frozenset({'nuts'}), items_add=frozenset({'beer'}),
confidence=1.0, lift=1.0)])]

Three major measures to validate Association Rules:

- Support
- Confidence
- Lift

Suppose a record of 1 thousand customer transactions. Consider two items e.g. burgers and ketchup. Out of one thousand transactions, 100 contain ketchup while 150 contain a burger. Out of 150 transactions where a burger is purchased, 50 transactions contain ketchup as well. Using this data, Find the support, confidence, and lift.

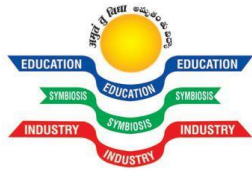
Support:

$\text{Support}(B) = (\text{Transactions containing } (B)) / (\text{Total Transactions})$

For instance if out of 1000 transactions, 100 transactions contain Ketchup then the support for item Ketchup can be calculated as:

$\text{Support}(\text{Ketchup}) = (\text{Transactions containing Ketchup}) / (\text{Total Transactions})$

$\text{Support}(\text{Ketchup}) = 100/1000 = 10\%$



Confidence:

Confidence refers to the likelihood that an item B is also bought if item A is bought. It can be calculated by finding the number of transactions where A and B are bought together, divided by total number of transactions where A is bought.

$$\text{Confidence}(A \rightarrow B) = (\text{Transactions containing both (A and B)}) / (\text{Transactions containing A})$$

A total of 50 transactions where Burger and Ketchup were bought together. While in 150 transactions, burgers are bought. Then we can find likelihood of buying ketchup when a burger is bought can be represented as confidence of Burger $\rightarrow$ Ketchup and can be mathematically written as:

$$\text{Confidence (Burger} \rightarrow \text{Ketchup)} = (\text{Transactions containing both (Burger and Ketchup)}) / (\text{Transactions containing A})$$

$$\text{Confidence(Burger} \rightarrow \text{Ketchup)} = 50/150 = 33.3\%$$

Lift :

Lift (A $\rightarrow$ B) refers to the increase in the ratio of sale of B when A is sold. Lift(A $\rightarrow$ B) can be calculated by dividing Confidence(A $\rightarrow$ B) divided by Support(B). Mathematically it can be represented as:

$$\text{Lift (A} \rightarrow \text{B)} = (\text{Confidence (A} \rightarrow \text{B)}) / (\text{Support (B)})$$

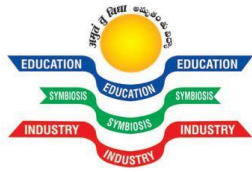
In Burger and Ketchup problem, the Lift (Burger $\rightarrow$ Ketchup) can be calculated as:

$$\text{Lift (Burger} \rightarrow \text{Ketchup)} = (\text{Confidence (Burger} \rightarrow \text{Ketchup)}) / (\text{Support (Ketchup)})$$

$$\text{Lift(Burger} \rightarrow \text{Ketchup)} = 33.3/10 = 3.33$$

- a) Display top 5 rows of data

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from apyori import apriori
store_data = pd.read_csv("D:/datasets/store_data.csv")
print(store_data.head())
print('Structure of store data\n',str(store_data))
```



- b) Find the rules with min_confidence : .2, min_support= 0.0045, min_lift=3, min_length=2

Let's suppose that we want rules for only those items that are purchased at least 5 times a day, or $7 \times 5 = 35$ times in one week, since our dataset is for a one-week time period.

The support for those items can be calculated as $35/7500 = 0.0045$.

The minimum confidence for the rules is 20% or 0.2.

Similarly, the value for lift as 3 and finally min_length is 2 since at least two products should exist in every rule.

#Converting data frame to list

```
records = []
```

```
for i in range(0, 7500):
```

```
records.append([str(store_data.values[i,j]) for j in range(0, 20)])
```

```
#Generating association rules using apriori()
```

```
#association_rules = apriori(records, min_support=0.0045, min_confidence=0.2, min_lift=3,  
min_length=2)
```

```
association_rules = apriori(records, min_support=0.0045, min_confidence=0.2, min_lift=3,  
min_length=5)
```

```
association_results = list(association_rules)
```

```
print(len(association_results))
```

```
print(association_results[0])
```

```
for item in association_rules:
```

```
# first index of the inner list
```

```
# Contains base item and add item
```

```
pair = item[0]
```

```
items = [x for x in pair]
```

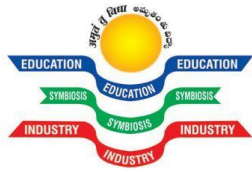
```
print("Rule: " + items[0] + " -> " + items[1])
```

```
#second index of the inner list
```

```
print("Support: " + str(item[1]))
```

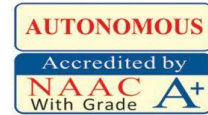
```
#third index of the list located at 0th
```

```
#of the third index of the inner list
```



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```
print("Confidence: " + str(item[2][0][2]))  
print("Lift: " + str(item[2][0][3])) print("=====")
```

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